ST. ANDREWS INSTITUTE OF TECHNOLOGY & MANAGEMENT
Gurgaon-Delhi(NCR)

BCA-103: MATHEMATICS (1^{st} Semester)

Unit 1

SETS

Introduction

The concept of set serves as a fundamental part of the present day mathematics. Today this concept is being used in almost every branch of mathematics. Sets are used to define the concepts of relations and functions. The study of geometry, sequences, probability, etc. requires the knowledge of sets.

In everyday life, we often speak of collections of objects of a particular kind, such as, a pack of cards, a crowd of people, a cricket team, etc. In mathematics also, we come across collections which we will study in further sections.

Definition 0.1. *A set is a well defined collection of objects or elements.*

Well-defined means, it must be absolutely clear that which object belongs to the set and which does not.

Example 1. *Examples of Set:*

1. *The collection of first five natural numbers.*
2. *The collection of vowels of the English Alphabet.*
3. *The collection of whole numbers between 20 and 25.*
4. *The collection of natural numbers between 30 and 35.*
5. *The collection of letters that are there in the word "MANGO".*
6. *The collection of letters that are there in the word "MOTION".*
7. *The collection of planets which are there in our solar system.*
8. *The collection of shops which are there in Dehradun.*
9. *The collection of letter in the word "DICTIONARY".*

10. *The collection of all the students in your school.*
11. *The collection of all the students who took the science stream in your school.*

Example 2. *Examples of not a set:*

1. *The collection of the most talented boys in your school.*
2. *The collection of the most dangerous animals which are found in the forest.*
3. *The collection of the best football players in the world.*
4. *Collection of the best musicians in the world.*
5. *Collection of the best cricket players in the world.*
6. *The collection of the most talented artists in the world.*
7. *The collection of time is not a set.*
8. *The collection of temperature.*
9. *The collection of best restaurants near your house.*

10 *The collection of the most talented Journalists in the world.*

We give below a few more examples of sets used particularly in mathematics, viz.

- $\mathbb{N}$: the set of all natural numbers
- $\mathbb{Z}$: the set of all integers
- $\mathbb{Q}$: the set of all rational numbers
- $\mathbb{R}$: the set of real numbers
- $\mathbb{Z}^+$: the set of positive integers
- $\mathbb{Q}^+$: the set of positive rational numbers, and
- $\mathbb{R}^+$: the set of positive real numbers.

The symbols for the special sets given above will be referred to throughout this text.

There are two methods which are in common use to denote a set.

(i) **Roster form:** all the elements of a set are listed, the elements are being separated by commas and are enclosed within braces $\{\}$. For example, the set of all even positive integers less than 7 is described in roster form as $\{2, 4, 6\}$.

(a) The set of all natural numbers which divide 42 is represented as $\{1, 2, 3, 6, 7, 14, 21, 42\}$.

Note: In roster form, the order in which the elements are listed is immaterial. Thus, the above set can also be represented as $\{1, 3, 7, 21, 2, 6, 14, 42\}$.

Note: It may be noted that while writing the set in roster form an element is not generally repeated, i.e., all the elements are taken as distinct. For example, the set of letters forming the word 'SCHOOL' is $\{S, C, H, O, L\}$ or $\{H, O, L, C, S\}$. Here,the order of listing elements has no relevance.

(ii) **Set-Builder Form**: all the elements of a set possess a single common property which is not possessed by any element outside the set. For example, in the set $\}a, e, i, o, u\}$, all the elements possess a common property, namely, each of them is a vowel in the English alphabet, and no other letter possess this property. Denoting this set by V, we write

$$V = \{x : x\text{is a vowel in English alphabet}\}.$$

(a) The set S of all elements x which have the property $P(x)$

$$S = \{x : P(x)\}$$

(b) The set B of natural numbers

$$B = \{n : n \in \mathbf{N}\}$$

Definition 0.2 (Null Set/ Empty Set/ Void Set:)**.** *A set having no element. It is generally denoted by the Danish letter ϕ or $\{\ \}$.*

We give below a few examples of empty sets.

Example 3. *(i) Let $A = \{x : 1 < x < 2, x \text{ is a natural number}\}$. Then A is the empty set, because there is no natural number between 1 and 2.*

(ii) $B = \{x : x^2 - 2 = 0 and x is rational number\}$. Then B is the empty set because the equation x^2 - 2 = 0 is not satisfied by any rational value of x.

(iii) $C = \{x : x is an even prime number greater than 2\}$. Then C is the empty set, because 2 is the only even prime number.

(iv) $D = \{x : x^2 = 4, x is odd\}$. Then D is the empty set, because the equation $x^2 = 4$ is not satisfied by any odd value of x.

Equal Sets

Definition 0.3. *Two sets are said to be equal when they consist of exactly the same elements.*

Example 4. *Examples of equal sets*

$$\{a, b, c\} = \{b, a, c\}$$

$$\{4, 5, 6, 7, 8, 9\} = \{n : 3 < n < 10, n \in N\}$$

Note: A set does not change if one or more elements of the set are repeated. For example, the sets $A = \{1, 2, 3\}$ and $B = \{2, 2, 1, 3, 3\}$ are equal, since each element of A is in B and vice-versa. That is why we generally do not repeat any element in describing a set.

Finite Set and Infinite set

Definition 0.4. *Finite sets are sets having a finite/countable number of members. Finite sets are also known as countable sets, as they can be counted.*

Example 5. *Examples of finite set*

- $P = \{0, 3, 6, 9, ..., 99\}$
- $Q = \{a : a \text{ is an integer}, 1 < a < 10\}$
- *A set of all English Alphabet (because it is countable).*

Cardinality of Finite Set

If 'a' represents the number of elements of set A, then the cardinality of a finite set is $n(A) = a$. The cardinality of a finite set is a natural number or possibly 0.

Example 6. *Find the cardinality of given sets*

1. *Cardinality of set A of all english alphabets is $n(A) = 26$.*
2. *A set containing the months in a year will have a cardinality of 12.*
3. *S = Set of the number of people living in India is a non-empty finite set.*

Can we say that an empty set is a finite set?

Let's learn what is an empty set first. An empty set is a set with no elements and can be represented as {} and shows that it has no element. P = {} or ϕ. As the finite set has a countable number of elements and the empty set has zero elements so, it is a definite number of elements. So, with a cardinality of zero, an empty set is a finite set.

Definition 0.5. *If a set is not finite, it is called an infinite set because the number of elements in that set is not countable. Thus, infinite sets are also known as uncountable sets.*

Example 7. *Examples of infinite sets*

1. *A set of all whole numbers, $W = \{0, 1, 2, 3, 4, ...\}$*
2. *A set of all points on a line*
3. *The set of all integers*

Cardinality of Infinite Sets

The cardinality of an infinite set is $n(A) = \infty$ as the number of elements is unlimited in it.

Note: All infinite sets cannot be described in the roster form. For example, the set of real numbers cannot be described in this form, because the elements of this set do not follow any particular pattern

Universal Set

Definition 0.6. *A universal set is a set which contains all the elements or objects of other sets, including its own elements. It is usually denoted by the symbol 'U'.*

Suppose Set A consists of all even numbers such that, $A = \{2, 4, 6, 8, 10, ...\}$ and set B consists of all odd numbers, such that, $B = \{1, 3, 5, 7, 9, ...\}$. The universal set U consists of all natural numbers, such that, $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...\}$.

Example 8. *Let us say, there are three sets named as A, B and C. The elements of all sets A, B and C is defined as: $A = \{1, 3, 6, 8\}$, $B = \{2, 3, 4, 5\}$ and $C = \{5, 8, 9\}$. Find the universal set for all the three sets A, B and C.*
Answer: *By the definition we know, the universal set includes all the elements of the given sets. Therefore, Universal set for sets A, B and C will be, $U = \{1, 2, 3, 4, 5, 6, 8, 9\}$*

Example 9. *If $C = \{2, 4, 6\}$, $A = \{1, 2, 3\}$, $T = \{6, 8, 9\}$. Then $U = \{1, 2, 3, 4, 6, 8, 9\}$*

Notation

$$\forall,\ \exists,\ \Rightarrow,\ \Leftrightarrow,\ \sim,\ \vee,\ \wedge$$

These symbols borrowed from mathematical logic help in a neat and brief exposition of the subject and so we shall describe them brief here.

(i) $\forall$ stands for 'for all' or 'for every'.
The statement $x < y$, $\forall\, x \in S$ means x is less than y for all members of S.

(ii) $\exists$ stands for 'there exists'.

(iii) $\Rightarrow$ stands for 'implies that'. $\Leftrightarrow$ stand for both ways implication.

(iv) $\vee$ stands for 'or'

(v) $\wedge$ stands for 'and'.

(vi) Negation $\sim$ stands for 'not.'

Subsets

Definition 0.7. *If A and B are two sets such that each member of A is also a member of B, i.e., $x \in A \Rightarrow x \in B$, then A is called a* ***subset*** *of B and we write $A \subseteq B$.*

Example 10. *Examples of subset*

- $A = \{1, 2, 3\}$ *is a subset of* $B = \{1, 2, 3, 4, 10\}$
- $A = \{p, q, r\}$ *is a subset of* $B =$ *set of all alphabets*
- $A =$ *set of all even numbers is a subset of* $B =$ *set of all integers*

Note: Every set is a subset of itself and also the empty set (ϕ) is also a subset of every set.

There are 2 types of subsets:

- Proper subset (The symbol used for this is $\subset$)
- Improper subset (The symbol used for this is $\subseteq$)

Proper subset

If a set contains only a few or no elements of another set, then it is the proper subset of the given set.

Proper Subset Formula

Another formula to calculate the number of proper subsets of a set is $2^n - 1$. Here, 'n' is the number of elements in a set.

Example 11. *If a set A = x, y, then calculate its number of proper subsets and write them.*
Solution:*Given that, $A = \{x, y\}$. Number of elements in set $A = 2$. Using proper subsets formula: $2^n - 1 = 2^2 - 1 =$ 4 - 1 = 3. Thus, the number of proper subsets of the set A = 3. Proper Subsets = ($\{\}$, $\{a\}$, $\{b\}$)*

Improper subset

If a set contains all elements of another set, it is called an improper subset of the original set.

Power Set

A set with the collection of all the subsets is known as Power Set.

Power Set Formula

If B has n elements, then $P(B)$ has 2^n, which means power sets will have certain elements.

Example 12. *If set B has $\{0,1\}$ elements than its power set will be: $2^n = 2^2 = 4$, $P(B) = \{\}, \{0\}, \{1\}, \{0,1\}$*

Question: Find the proper and improper subsets of $B = \{2,4,6\}$.
Solution: Given set $B = \{2,4,6\}$. Subsets: $\{\}, \{2\}, \{4\}, \{6\}, \{2,4\}, \{4,6\}, \{2,6\}, \{2,4,6\}$. Here, Proper subsets = $\{\}, \{2\}, \{4\}, \{6\}, \{2,4\}, \{4,6\}, and \{2,6\}$. Improper subset= $\{2,4,6\}$.

Operations of set

Union, Intersection and Complement

Definition 0.8. *Let X and Y be two sets.*

1. *The union of X and Y, denoted by $X \cup Y$, is the set that consists of all elements of X and also all elements of Y. More specifically, $X \cup Y = \{x | x \in X \, or \, x \in Y\}$.*
2. *The intersection of X and Y , denoted by $X \cap Y$, is the set of all common elements of X and Y. More specifically, $X \cap Y = \{x | x \in X \, and \, x \in Y\}$.*
3. *The sets X and Y are said to be disjoint if $X \cap Y = \phi$.*

Definition 0.9. *The complement of a set A contains everything that is not in the set A. The complement is notated A', or A^c, or sometimes $\sim A$.*

Some Properties of Complement Sets

1. **De Morgan's laws:** If A and B are any two subsets of the universal set U, then
   (i) $(A \cup B)' = A' \cap B'$.
   Proof:
   We need to prove, $(A \cup B)' = A' \cap B'$.

   Let $X = (A \cup B)'$ and $Y = A' \cap B'$

   Let y be any element of X, then $y \in X \Rightarrow y \in (A \cup B)'$.

   $\Rightarrow y \notin (A \cup B)$.

   $y \notin A$ or $y \notin B$

   $\Rightarrow y \in A'$ and $y \in B'$

   $\Rightarrow y \in A' \cap B'$

$\Rightarrow y \in Y$

$\therefore X \subset Y\ .\ .\ .$ (i)

Again, let z be any element of Y, then $z \in Y \Rightarrow z \in A' \cap B'$

$\Rightarrow z \in A'$ and $z \in B'$

$\Rightarrow z \notin A$ or $z \notin B$

$\Rightarrow z \notin (A \cup B)$

$\Rightarrow z \in (A \cup B)'$

$\Rightarrow z \in X$

$\therefore Y \subset X\ .\ .\ .$ (ii)

From (i) and (ii) X = Y

Hence $(A \cup B)' = A' \cap B'$.

(ii) $(A \cap B)' = A' \cup B'$.

Proof:

Let $X = (A \cap B)'$ and $Y = A' \cup B'$.

Let p be any element of X, then $p \in X \Rightarrow p \in (A \cap B)'$.

$\Rightarrow p \notin (A \cap B)$.

$\Rightarrow p \notin A$ and $p \notin B$.

$\Rightarrow p \in A'$ or $p \in B'$.

$\Rightarrow p \in A' \cup B'$.

$\Rightarrow p \in Y$.

$\therefore X \subset Y$ ....(i)

Again, let q be any element of Y, then $q \in Y \Rightarrow q \in A' \cup B'$.

$\Rightarrow q \in A'$ or $q \in B'$.

$\Rightarrow q \notin A$ and $q \notin B$.

$\Rightarrow q \notin (A \cap B)$.

$\Rightarrow q \in (A \cap B)'$.

$\Rightarrow q \in X$.

$\therefore Y \subset X$ ....(ii)

From (i) and (ii) X = Y

$(A \cap B)' = A' \cup B'$.

2. **Complement laws:**
   (i) $A \cup A' = U$
   (ii) $A \cap A' = \phi$
3. Law of double complementation : $(A')' = A$
4. Laws of empty set and universal set $\phi' = U$ and $U' = \phi$.

Example 13. *Consider the sets:*
$A = \{red, green, blue\}$, $B = \{red, yellow, orange\}$ *and* $C = \{red, orange, yellow, green, blue, purple\}$.
Find the following:

(a) Find $A \cup B$.

(b) Find $A \cap B$.

(c) Find $A^c \cap B$.

Answers:

(a) The union contains all the elements in either set: $A \cup B = \{red, green, blue, yellow, orange\}$. *Notice we only list red once.*

(b) The intersection contains all the elements in both sets: $A \cap B = \{red\}$.

(c) Here we're looking for all the elements that are not in set A *and are also in* C. $A^c \cap C = \{orange, yellow, purple\}$

Example 14. *1. Let* $A = \{1, 2, 4, 18\}$ *and* $B = \{x : x$ *is an integer,* $0 < x \leq 5\}$. *Then,* $A \cup B = \{1, 2, 3, 4, 5, 18\}$ *and* $A \cap B = \{1, 2, 4\}$.

2. Let $S = \{x \in R : 0 \leq x \leq 1\}$ *and* $T = \{x \leq R : 0.5 \leq x < 7\}$. *Then,* $S \cup T = \{x \in R : 0 \leq x < 7\}$ *and* $S \cap T = \{x \in R : 0.5 \leq x \leq 1\}$.

3. *Let* $X = \{\{b,c\},\{\{b\},\{c\}\},b\}$ *and* $Y = \{a,b,c\}$. *Then* $X \cap Y = \{b\}$ *and* $X \cup Y = \{a,b,c,\{b,c\},\{\{b\},\{c\}\}\}$.

We now state a few properties related to the union and intersection of sets.

Lemma 0.10. *Let R, S and T be sets. Then,*

1. a) $S \cup T = T \cup S$ *and* $S \cap T = T \cap S$ *(union and intersection are commutative operations).*
   b) $R \cup (S \cup T) = (R \cup S) \cup T$ *and* $R \cap (S \cap T) = (R \cap S) \cap T$ *(union and intersection are associative operations).*
   c) $S \subseteq S \cup T$, $T \subseteq S \cup T$.
   d) $S \cap T \subseteq S$, $S \cap T \subseteq T$.
   e) $S \cup \phi = S$, $S \cap \phi = \phi$.
   f) $S \cup S = S \cap S = S$.

2. *Distributive laws (combines union and intersection):*

   a) $R \cup (S \cap T) = (R \cup S) \cap (R \cup T)$ *(union distributes over intersection).*
   Proof:
   Let x be an arbitary element of $A \cup (B \cap C)$. *Then,* $x \in A \cup (B \cap C)$
   $\implies x \in A$ *or* $x \in (B \cap C)$
   $\implies x \in A$ *or* $x \in B$ *and* $x \in C$
   $\implies (x \in A$ *or* $x \in B)$ *and* $(x \in Aorx \in C)$
   [$\therefore$ *'or' distributes 'and'*]
   $\implies x \in (A \cup B)$ *and* $x \in (A \cup C)$.
   $\implies x \in (A \cup B) \cap (A \cup C)$
   $\therefore A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$*......(i)*
   Again. let y be an arbitary element of $(A \cup B) \cap (A \cup C)$. *Then,* $y \in (A \cup B) \cap (A \cup C)$
   $\implies y \in (A \cup B)$ *and* $y \in (A \cup C)$
   $\implies (y \in A$ *or* $y \in B)$ *and* $(y \in A$ *or* $y \in C)$
   $\implies y \in A$ *or* $(y \in Bandy \in C)$
   [$\therefore$ *'or' distributes 'and'*]
   $y \in A$ *or* $y \in (B \cap C)$
   $\implies y \in A \cup (B \cap C)$
   $\therefore (A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$*......(ii).*
   From (i) and (ii), we get
   $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

   b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ *(intersection distributes over union).*
   Proof:
   Let x be an arbitrary element of $A \cap (B \cup C)$. *Then,* $x \in A \cap (B \cup C)$

   $\implies x \in A$ *and* $x \in (B \cup C)$

   $\implies x \in A$ *and* $(x \in Borx \in C)$

$\implies (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)$

[$\therefore$ *'and' distributes 'or'*]

$\implies x \in (A \cap B) \text{ or } x \in (A \cap C)$

$\implies x \in (A \cap B) \cup (A \cap C)$

$\therefore A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$........*(i)*

Again, let y *be an arbitrary element of* $(A \cap B) \cup (A \cap C)$. *Then,* $y \in (A \cap B) \cup (A \cap C)$

$\implies y \in (A \cap B) \text{ or } y \in (A \cap C)$

$\implies (y \in A \text{ and } y \in B) \text{ or } (y \in A \text{ and } y \in C)$

$\implies y \in A \text{ and } (y \in B \text{ or } y \in C)$

[$\therefore$ *'and' distributes 'or'*]

$\implies y \in A \text{ and } y \in (B \cup C)$

$\implies y \in A \cap (B \cup C)$.

$\therefore (A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$..........*(ii)*

From (i) and (ii), we get

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Definition 0.11. *Let* X *and* Y *be two sets.*

1. *The set difference of* X *and* Y*, denoted by* $X \backslash Y$ *or* $X - Y$*, is defined by* $X - Y = \{x \in X : x \notin Y\}$.

2. *The set* $(X - Y) \cup (Y - X)$*, denoted by* $X \Delta Y$ *or* $A + B$*, is called the symmetric difference of* X *and* Y.

Example 15.

1. *Let* $A = \{1, 2, 4, 18\}$ *and* $B = \{x \in Z : 0 < x \leq 5\}$. *Then,* $A - B = \{18\}$, $B - A = \{3, 5\}$ *and* $A \Delta B = \{3, 5, 18\}$.

2. *Let* $S = \{x \in R : 0 \leq x \leq 1\}$ *and* $T = \{x \in R : 0.5 \leq x < 7\}$. *Then,* $S - T = \{x \in R : 0 \leq x < 0.5\}$ *and* $T - S = \{x \in R : 1 < x < 7\}$.

3. *Let* $X = \{\{b, c\}, \{\{b\}, \{c\}\}, b\}$ *and* $Y = \{a, b, c\}$. *Then* $X - Y = \{\{b, c\}, \{\{b\}, \{c\}\}\}$, $Y - X = \{a, c\}$ *and* $X \Delta Y = \{a, c, \{b, c\}, \{\{b\}, \{c\}\}\}$.

Cartesian Product

The Cartesian product of the sets A and B, is written as $A \times B$, is the set of all ordered pairs in which the first elements are in A and the second elements are in B. i.e. $A \times B = \{\langle x, y\rangle | x \in A \quad \text{and} \quad x \in B\}$

Example 16. *Let* $A = \{1, 2\}$*,* $B = \{a, b, c\}$*,* $C = \{\alpha, \beta\}$*, Then*

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

$$A \times C = \{(1, \alpha), (1, \beta), (2, \alpha), (2, \beta)\}$$

$$B \times C = \{(a, \alpha), (a, \beta), (b, \alpha), (b, \beta), (c, \alpha), (c, \beta)\}$$

Some properties of Cartesian Product

- $A \times B \neq B \times A$, i.e Cartesian product is not commutative.
- $A \times (B \times C) = \{(a, (b, c)) | a \in A \quad \text{and} \quad (b, c) \in B \times C\}$. Moreover, Cartesian product is not associative, i.e $(A \times B) \times C \neq A \times (B \times C)$.
- If A has n elements and B has m elements then $A \times B$ has nm elements.